Axolotl

1. What is Axolotl
2. Why config-driven matters
3. What Axolotl can do that HF alone cannot
4. Training methods
5. Dataset & scalability
6. Difference between core hf, unsloth, llamafactory, axolotl
7. Live Docker demo (Finetuning with minimum steps)

1. What is Axolotl?

Axolotl = config-driven (CLI-first) framework with Python extensibility
Open source
Free to use
Industry-oriented
built on Hugging Face ecosystem

2. Axolotl = CONFIG-DRIVEN TRAINING ENGINE

This is Axolotl's biggest strength
One YAML controls:

Dataset loading
Tokenization
Training
Evaluation
Quantization
Inference
"Unlike Hugging Face where code controls training, Axolotl flips it: YAML controls everything."

Why config-driven matters

Reproducibility
Faster Experiment (Research + Production both)
Automation & CI/CD Friendly

3. What Axolotl can do that HF alone cannot

Features:

- **Multiple Model Support**: Train various models like GPT-OSS, LLaMA, Mistral, Mixtral, Pythia, and many more models available on the Hugging Face Hub.
- **Multimodal Training**: Fine-tune vision-language models (VLMs) including LLaMA-Vision, Qwen2-VL, Pixtral, LLaVA, SmolVLM2, and audio models like Voxtral with image, video, and audio support.
- **Training Methods**: Full fine-tuning, LoRA, QLoRA, GPTQ, QAT, Preference Tuning (DPO, IPO, KTO, ORPO), RL (GRPO), and Reward Modelling (RM) / Process Reward Modelling (PRM).
- **Easy Configuration**: Re-use a single YAML configuration file across the full fine-tuning pipeline: dataset preprocessing, training, evaluation, quantization, and inference.
- **Performance Optimizations**: Multipacking, Flash Attention, Xformers, Flex Attention, Liger Kernel, Cut Cross Entropy, Sequence Parallelism (SP), LoRA optimizations, Multi-GPU training (FSDP1, FSDP2, DeepSpeed), Multi-node training (Torchrun, Ray), and many more!
- **Flexible Dataset Handling**: Load from local, HuggingFace, and cloud (S3, Azure, GCP, OCI) datasets.
- **Cloud Ready**: We ship Docker images and also PyPI packages for use on cloud platforms and local hardware.

4. Training Methods Supported

"Axolotl is not just LoRA — it's a full post-training platform."

Fine-tuning:
Full FT
LoRA / QLoRA
GPTQ
QAT
FP8

Preference / RL:
DPO, IPO, KTO, ORPO
GRPO
RM, PRM

5. Dataset Support

Local
Hugging Face
Cloud (S3, Azure, GCP, OCI)
Instruction, chat, preference, vision, audio

6. Diff between HF/Unsloth/LLaMA-Factory/Axolotl

#	Dimension	Hugging Face (HF)	Unsloth	LLaMA-Factory	Axolotl
1	Core Purpose	Base ML framework for model training & inference	Performance engine that accelerates HF	End-to-end fine-tuning platform with UI	Training orchestrator for scalable experiments
2	Performance (Speed & VRAM)	Baseline speed, high VRAM usage	2–3× faster, 50–80% less VRAM (verified)	Same as HF (no kernel optimization)	Same as HF (no kernel optimization)
3	GPU-level Optimizations	None (uses standard PyTorch ops)	Custom CUDA + Triton kernels, manual backprop	None	None
4	Workflow & Scale	Code-heavy, manual setup	Simple code, single-GPU focused	UI + CLI driven, structured but less scalable	YAML-driven, reproducible, multi-GPU, DeepSpeed/FSDP